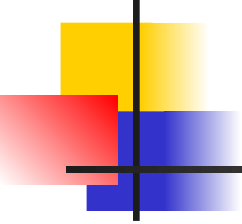


Cluster Analysis

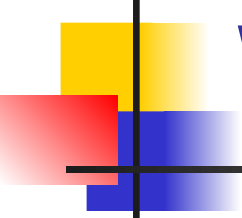
Dr. Bernard Chen

Troy University



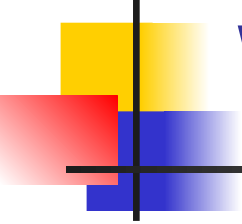
Supervised learning VS Unsupervised learning

- If the class label of each training tuple is provided, this process is known as **supervised learning**
- It contrasts with **unsupervised learning** (or clustering), in which the class label of each training tuple is **unknown**



What is Cluster Analysis?

- Cluster: a collection of data objects
 - Similar to one another within the same cluster
 - Dissimilar to the objects in other clusters
- Cluster analysis
 - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters



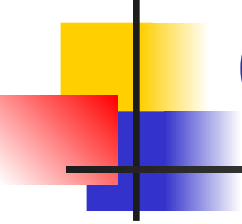
What is Cluster Analysis?

- Clustering analysis is an important human activity
- Early in childhood, we learn how to distinguish between **cats** and **dogs**
- **Unsupervised learning**: no predefined classes
- Typical applications
 - As a **stand-alone tool** to get insight into data distribution
 - As a **preprocessing step** for other algorithms



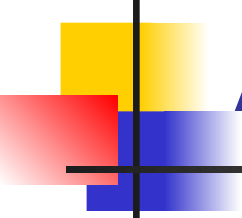
Clustering: Rich Applications and Multidisciplinary Efforts

- Pattern Recognition
- Spatial Data Analysis
 - Create thematic maps in GIS by clustering feature spaces
 - Detect spatial clusters or for other spatial mining tasks
- Image Processing
- Economic Science (especially market research)
- WWW
 - Document classification
 - Cluster Weblog data to discover groups of similar access patterns



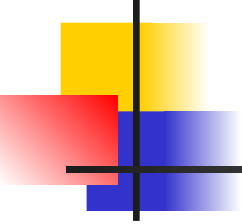
Quality: What Is Good Clustering?

- A good clustering method will produce high quality clusters with
 - high intra-class similarity
(Similar to one another within the same cluster)
 - low inter-class similarity
(Dissimilar to the objects in other clusters)
- The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns



Major Clustering Approaches

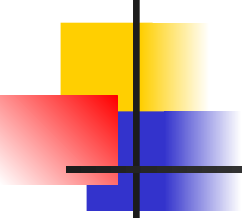
- Partitioning approach:
 - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: **k-means**, k-medoids, CLARANS
- Hierarchical approach:
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: **Hierarchical**, Diana, Agnes, BIRCH, ROCK, CAMELEON
- Density-based approach:
 - Based on connectivity and density functions
 - Typical methods: **DBSCAN**, OPTICS, DenClue



Partitioning Algorithms: Basic Concept

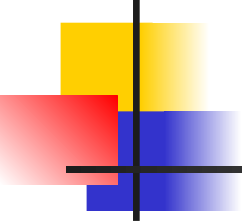
- Partitioning method: Construct a partition of a database ***D*** of ***n*** objects into a set of ***k*** clusters, s.t., min sum of squared distance

$$\sum_{m=1}^k \sum_{t_{mi} \in K_m} (C_m - t_{mi})^2$$



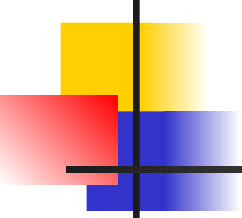
Partitioning Algorithms: Basic Concept

- Given a k , find a partition of k *clusters* that optimizes the chosen partitioning criterion
 - Global optimal: exhaustively enumerate all partitions
 - Heuristic methods: *k-means* and *k-medoids* algorithms
 - *k-means* (MacQueen'67): Each cluster is represented by the center of the cluster
 - *k-medoids* or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

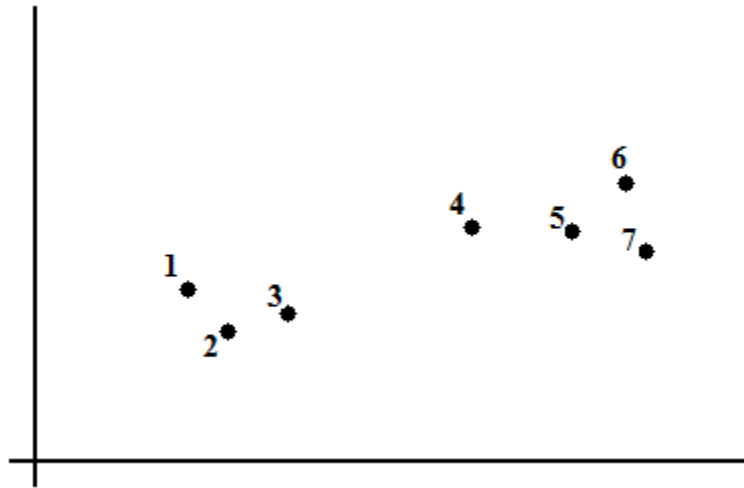


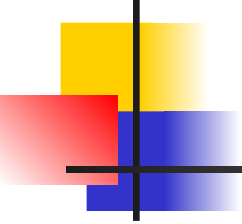
The *K-Means* Clustering Method

- Given k , the *k-means* algorithm is implemented in four steps:
 1. Partition objects into k nonempty subsets
 2. Compute seed points as the centroids of the clusters of the current partition (the centroid is the center, i.e., *mean point*, of the cluster)
 3. Assign each object to the cluster with the nearest seed point
 4. Go back to Step 2, stop when no more new assignment

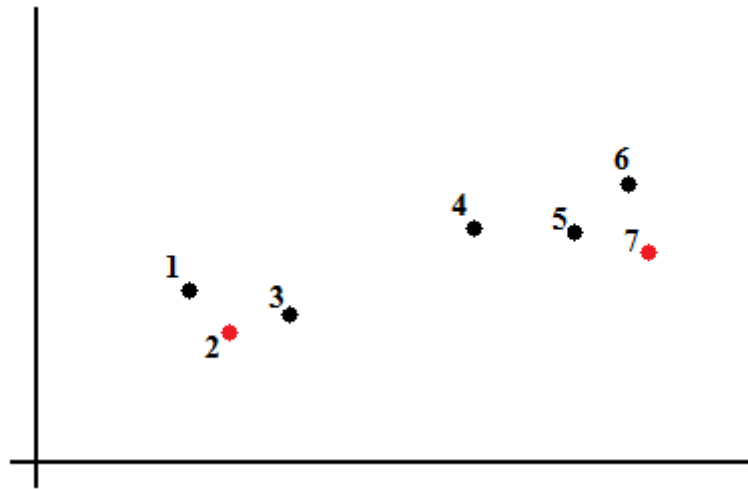


K-means Clustering

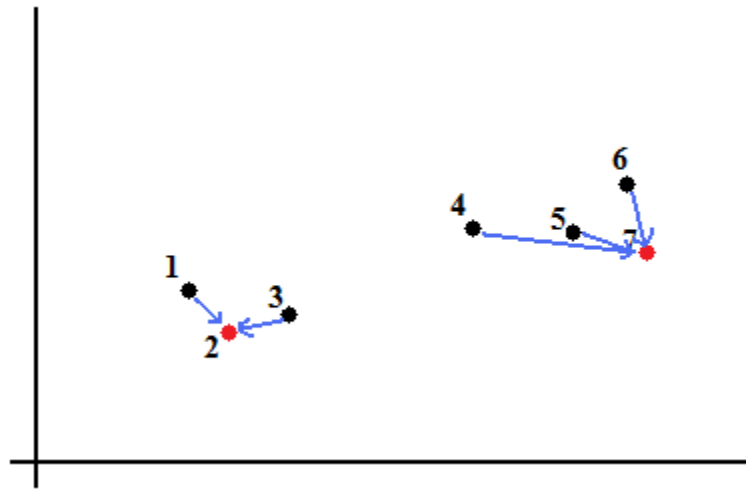


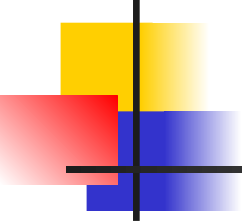


K-means Clustering

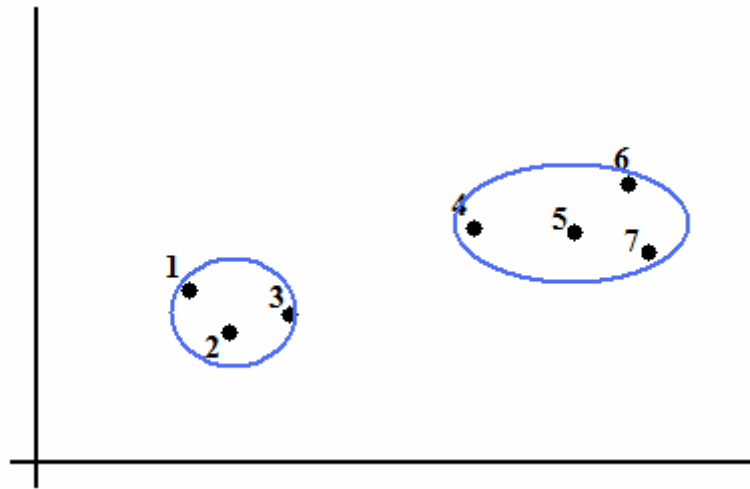


K-means Clustering

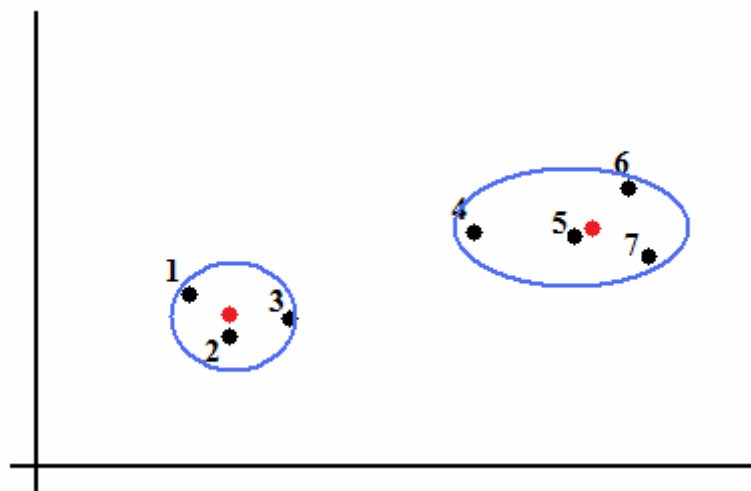




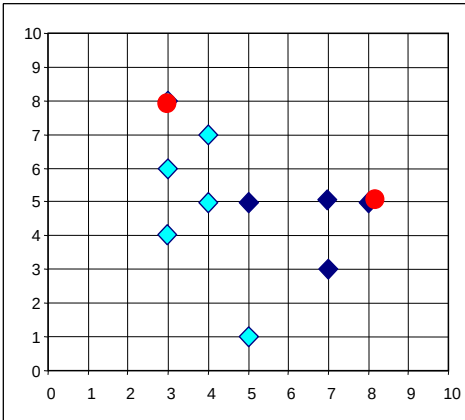
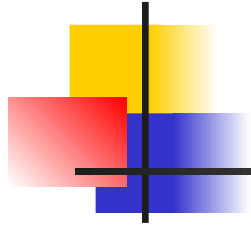
K-means Clustering



K-means Clustering



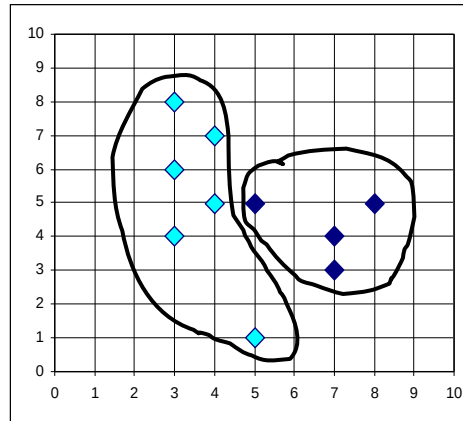
The *K-Means* Clustering Method



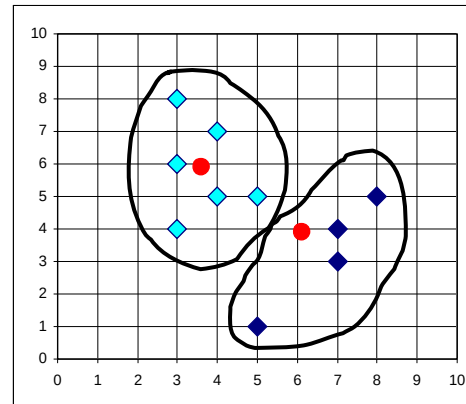
$K=2$

Arbitrarily choose
K object as initial
cluster center

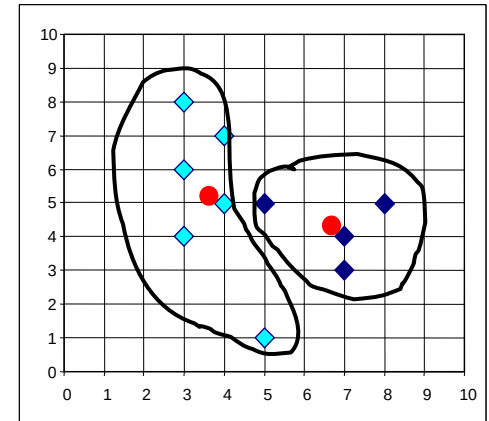
Assign each
objects to
most
similar
center



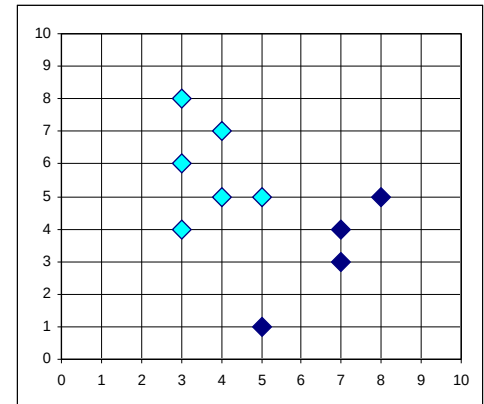
↑ reassign



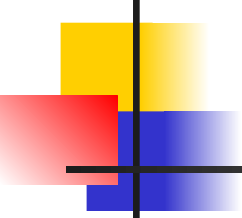
Update
the
cluster
means



↓ reassign

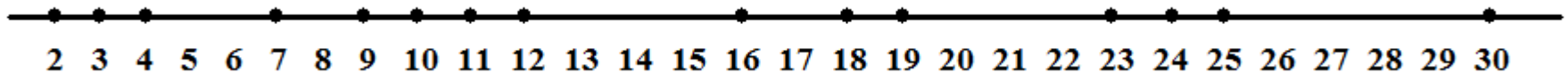


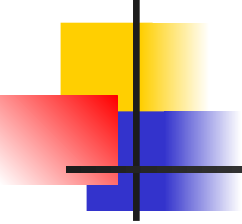
Update
the
cluster
means



Example

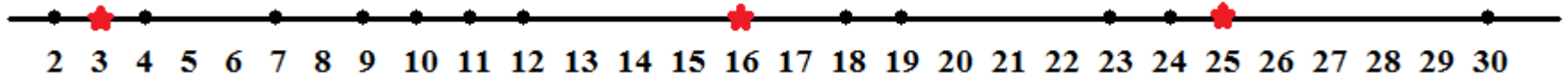
- Run K-means clustering with 3 clusters (initial centroids: 3, 16, 25) for at least 2 iterations

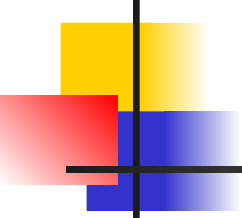




Example

- Run K-means clustering with 3 clusters (initial centroids: 3, 16, 25) for at least 2 iterations





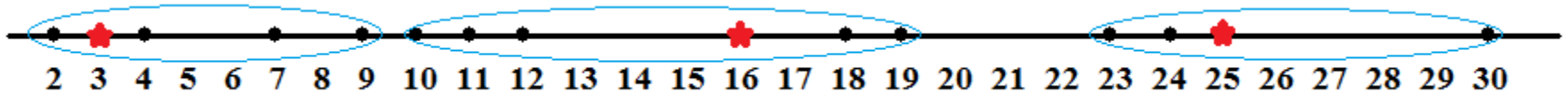
Example

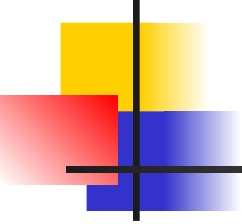
- Centroids:

3 - 2 3 4 7 9

16 - 10 11 12 16 18 19

25 - 23 24 25 30





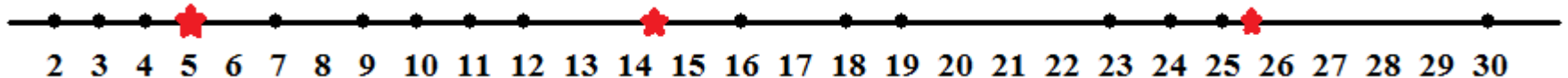
Example

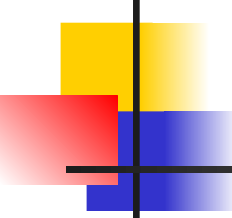
- Centroids:

3 – 2 3 4 7 9 new centroid: 5

16 – 10 11 12 16 18 19 new centroid: 14.33

25 – 23 24 25 30 new centroid: 25.5





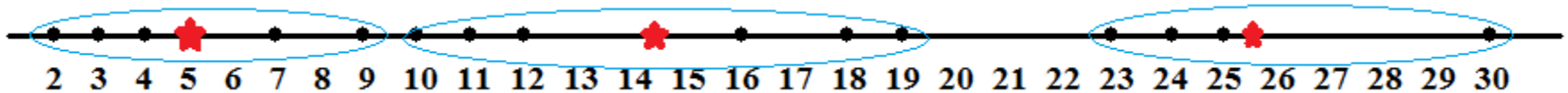
Example

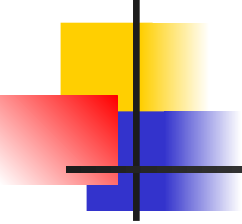
- Centroids:

5 – 2 3 4 7 9

14.33 – 10 11 12 16 18 19

25.5 – 23 24 25 30





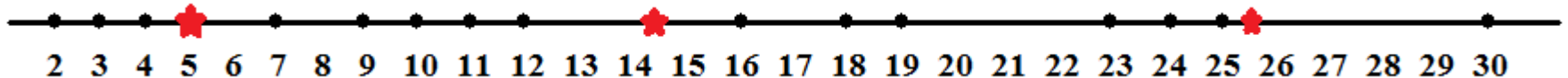
Example

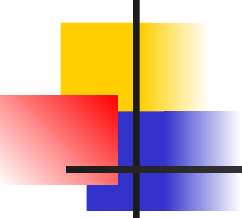
- Centroids:

5 – 2 3 4 7 9 new centroid: 5

14.33 – 10 11 12 16 18 19 new centroid: 14.33

25.5 – 23 24 25 30 new centroid: 25.5





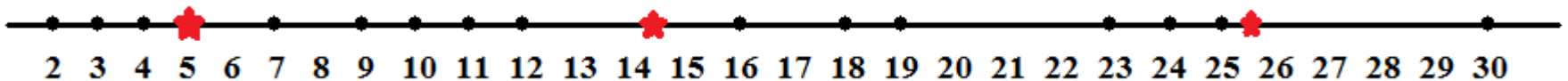
Example

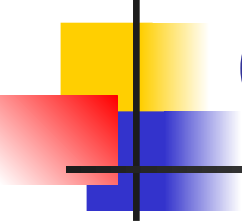
- Centroids:

5 – 2 3 4 7 9 new centroid: 5

14.33 – 10 11 12 16 18 19 new centroid: 14.33

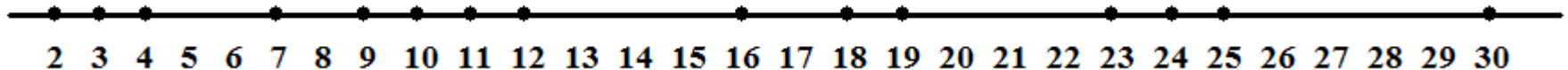
25.5 – 23 24 25 30 new centroid: 25.5

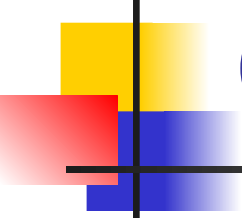




Class Assignment

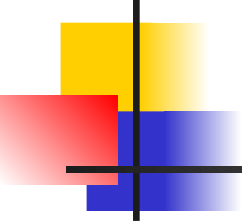
- Run K-means clustering with 3 clusters (initial centroids: 3, 12, 19) for at least 2 iterations





Look deeper into Clustering

- Three more things can be discussed in clustering:
 - Distance measure
 - Distance between clusters
 - Centroid location



Similarity and Dissimilarity Between Objects

- Distances are normally used to measure the similarity or dissimilarity between two data objects
- Some popular ones include: *Minkowski distance*:

$$d(i, j) = \sqrt[q]{(|x_{i1} - x_{j1}|^q + |x_{i2} - x_{j2}|^q + \dots + |x_{ip} - x_{jp}|^q)}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and q is a positive integer

- If $q = 1$, d is Manhattan distance

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

Similarity and Dissimilarity Between Objects (Cont.)

- If $q = 2$, d is Euclidean distance:

$$d(i, j) = \sqrt{(|x_{i_1} - x_{j_1}|^2 + |x_{i_2} - x_{j_2}|^2 + \dots + |x_{i_p} - x_{j_p}|^2)}$$

- Also, one can use weighted distance, parametric Pearson correlation, or other dissimilarity measures